

ITA0608 – Machine Learning Lab Practicals

EXP 04:

Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

Aim:

To build an Artificial Neural Network (ANN) by implementing the Backpropagation Algorithm and test the model using an appropriate dataset.

Algorithm:

- Input data is fed into the network (Forward Propagation).
- The network computes the output.
- The error between the predicted and actual output is calculated.
- The error is propagated backward through the network.
- The weights are updated using gradient descent to minimize the error.
- This process repeats for multiple epochs until the network learns the pattern.

Program:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, classification_report

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42 )

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = MLPClassifier(
    hidden_layer_sizes=(5,)
```

```

activation='relu',
solver='adam',
max_iter=1000,
random_state=42 )
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy of ANN Model:", round(accuracy * 100, 2), "%")
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
new_sample = [[5.1, 3.5, 1.4, 0.2]]
new_sample_scaled = scaler.transform(new_sample)
prediction = model.predict(new_sample_scaled)
print("\nNew Sample:", new_sample[0])
print("Predicted Class:", iris.target_names[prediction][0])

```

Output:

```

... Accuracy of ANN Model: 100.0 %

Classification Report:

```

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	19
versicolor	1.00	1.00	1.00	13
virginica	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

```

New Sample: [5.1, 3.5, 1.4, 0.2]
Predicted Class: setosa
/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:691: ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
warnings.warn(

```

Result:

The Artificial Neural Network (ANN) was successfully built and trained using the Backpropagation Algorithm in Python with the Iris dataset.

